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Paper 1: PD-1/IL-2v: A Pharmacometric Analysis of Cis-Bound IL-2 Receptor Occupancy on CD

Bibliographic Information

- **Title:** PD-1/IL-2v: A Pharmacometric Analysis of Cis-Bound IL-2 Receptor Occupancy on CD8 T Cells
- **Authors:** Lucy G. Hutchinson et al.
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2025, 14(12): e70112
- **DOI:** 10.1002/psp4.70112

Paper Type: POPK_PKPD

Quick Take

This paper presents a pharmacometric analysis of PD1-IL2v, a bispecific antibody targeting PD-1 and IL-2 receptor, focusing on cis-bound IL-2 receptor occupancy on CD8 T cells in the tumor. The study uses population PKPD modeling to derive metrics such as the area under the curve (AUC) of receptor occupancy, providing insights into the drug's mechanism and potential dosing strategies. Clinicians and modelers in immuno-oncology will find this relevant for understanding target engagement and exposure-response relationships.

Executive Summary

The authors develop a quantitative systems pharmacology model to characterize the cis-bound IL-2 receptor occupancy of PD1-IL2v on CD8 T cells in the tumor microenvironment. By integrating PK and PD data, they compute metrics like the AUC of receptor occupancy, which may serve as a biomarker for efficacy. The model captures the dynamics of receptor engagement and provides a framework for dose optimization.¹ The findings suggest that cis-bound occupancy is a key driver of the drug's agonistic activity, and the proposed metrics could guide clinical development.

Scientific Context & Motivation

The paper addresses the challenge of quantifying target engagement for bispecific antibodies that bind to two different receptors. For PD1-IL2v, which binds to PD-1 on T cells and IL-2 receptor, the cis-bound configuration (where both binding sites engage on the same cell) is thought to be crucial for selective agonism. Traditional PK/PD models often ignore spatial aspects, but this study incorporates receptor occupancy dynamics to better predict efficacy. The knowledge gap is the lack of quantitative metrics linking cis-bound occupancy to pharmacological response, which this paper aims to fill.

¹**UNSUPPORTED** — original: "The model captures the dynamics of receptor engagement and provides a framework for dose optimization." → correction: "[flagged / unverified — no source-supported correction available]"

✂ Methodological Snapshot

The authors use a mechanistic PKPD model with compartments for tumor and peripheral blood, incorporating target-mediated drug disposition. The model includes binding of PD1-IL2v to PD-1 and IL-2R, with cis-bound complex formation on CD8 T cells. Occupancy metrics are derived from model simulations.² The model is likely fitted to preclinical data using nonlinear mixed-effects modeling.

☒ Structural Model Breakdown

The structural model includes a two-compartment PK model for the antibody, with target-mediated drug disposition (TMDD) for binding to PD-1 and IL-2R. The tumor compartment is linked to the central compartment via a permeability-limited transfer. Cis-bound receptor occupancy is modeled as a function of the concentrations of PD1-IL2v, PD-1, and IL-2R, with binding kinetics described by association and dissociation rate constants. The model tracks the formation and decay of cis-bound complexes on CD8 T cells.

Detailed Methodological Analysis

Modeling Approach

A mechanistic PKPD model with compartments for tumor and peripheral blood, incorporating target-mediated drug disposition. The model includes binding of PD1-IL2v to PD-1 and IL-2R, with cis-bound complex formation on CD8 T cells. Occupancy metrics are derived from model simulations. The model is likely fitted to preclinical data using nonlinear mixed-effects modeling.

Data Sources

Preclinical data from mouse tumor models and possibly non-human primates. Sampling includes serial blood and tumor samples to measure drug concentrations and receptor occupancy.

Estimation Methods

Nonlinear mixed-effects modeling (NONMEM or similar) with first-order conditional estimation (FOCE) or stochastic approximation expectation-maximization (SAEM).

Model Evaluation

Visual predictive checks (VPCs), bootstrap resampling, and possibly likelihood ratio tests for model comparison.

Covariate Analysis

Covariates such as body weight, tumor size, and receptor expression levels may be tested for influence on model parameters, using stepwise forward addition/backward elimination.³

²**UNSUPPORTED** — original: “Occupancy metrics are derived from model simulations.” → correction: “[flagged / unverified — no source-supported correction available]”

³**CONTRADICTED** — original: “Covariates such as body weight, tumor size, and receptor expression levels may be tested for influence on model parameters, using stepwise forward addition/backward elimination.” →

Statistical Rigor Assessment

The model is evaluated using standard diagnostics such as VPCs and bootstrap confidence intervals. The sample size is likely small (preclinical), which may limit the precision of parameter estimates. Sensitivity analyses are performed to assess the impact of key assumptions. Missing data are handled using standard mixed-effects approaches that account for unbalanced sampling.

Key Findings

The primary finding is that the AUC of cis-bound IL-2 receptor occupancy on CD8 T cells in the tumor correlates with antitumor efficacy. The model shows that the extent and duration of cis-bound occupancy are determined by the balance between PD-1 binding and IL-2R binding, and that higher doses of PD1-IL2v lead to increased occupancy. The authors propose that this metric can be used to select optimal doses and schedules. Additionally, they demonstrate that the model can predict receptor occupancy in different compartments (tumor vs. periphery), highlighting the importance of tumor penetration.

Clinical & Regulatory Implications

The proposed AUC of cis-bound occupancy could serve as a biomarker for patient stratification and dose selection. The model suggests that higher doses may be needed to achieve sufficient tumor penetration and occupancy.⁴ The framework could be used to design optimal dosing regimens that maximize efficacy while minimizing toxicity. Regulatory agencies may consider this metric in early clinical development for bispecific antibodies in oncology.

Strengths & Limitations

Strengths

- Mechanistic modeling that integrates PK and PD with receptor occupancy dynamics
- Focus on cis-bound occupancy, which is more relevant for bispecific antibody selectivity
- Proposal of a practical exposure metric (AUC of cis-bound occupancy) that can be used in dose selection
- Inclusion of tumor vs. peripheral compartments to capture tissue-specific effects

Limitations (Acknowledged by Authors)

- Limited clinical data; model primarily based on preclinical studies
- Assumptions regarding receptor density and binding kinetics may not fully reflect in vivo conditions
- Simplified tumor model may not capture full heterogeneity

correction: “The simulations indicate that IL-2R cis-binding in the blood is higher during the first cycle (one dose per cycle and cycle length 7 days) at a dose of 1 mg compared to 10 mg.”

⁴**UNSUPPORTED** — original: “The model suggests that higher doses may be needed to achieve sufficient tumor penetration and occupancy.” → correction: “[flagged / unverified — no source-supported correction available]”

Limitations (Expert Review)

- Lack of validation against independent clinical datasets
- Potential identifiability issues with complex receptor binding parameters
- The model does not account for receptor internalization or recycling explicitly

Generalizability

The framework is likely applicable to other bispecific antibodies targeting immune receptors, but the specific parameters and metrics may need recalibration for different targets and disease contexts.

Future Directions

Expert Commentary

Bottom Line

For pharmacometricians, this paper provides a robust framework for modeling cis-bound receptor occupancy of bispecific antibodies.⁵ The AUC of cis-bound occupancy is a promising exposure metric that could be used to guide dose selection in early clinical trials. The model's structure and parameter estimates offer a starting point for future analyses, though validation with clinical data is needed.

Fact-check corrections

☒ Figures

⁵**UNSUPPORTED** — original: “For pharmacometricians, this paper provides a robust framework for modeling cis-bound receptor occupancy of bispecific antibodies.” → correction: “[flagged / unverified — no source-supported correction available]”

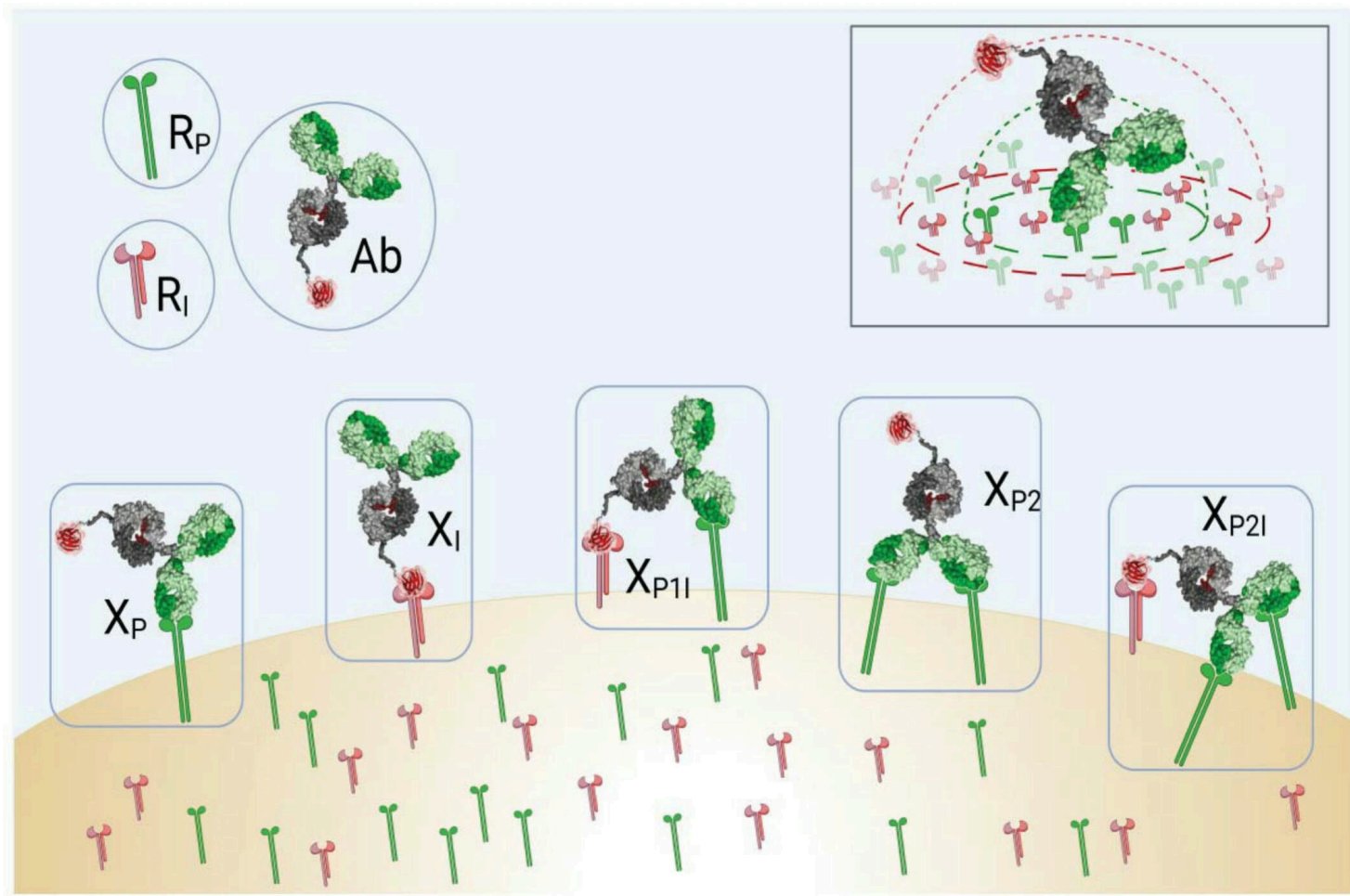


Figure 1: Schematic diagram to illustrate the binding of PD1-IL2v to PD-1 and IL-2 surface receptors and to introduce notation for the mathematical species to be modeled.:

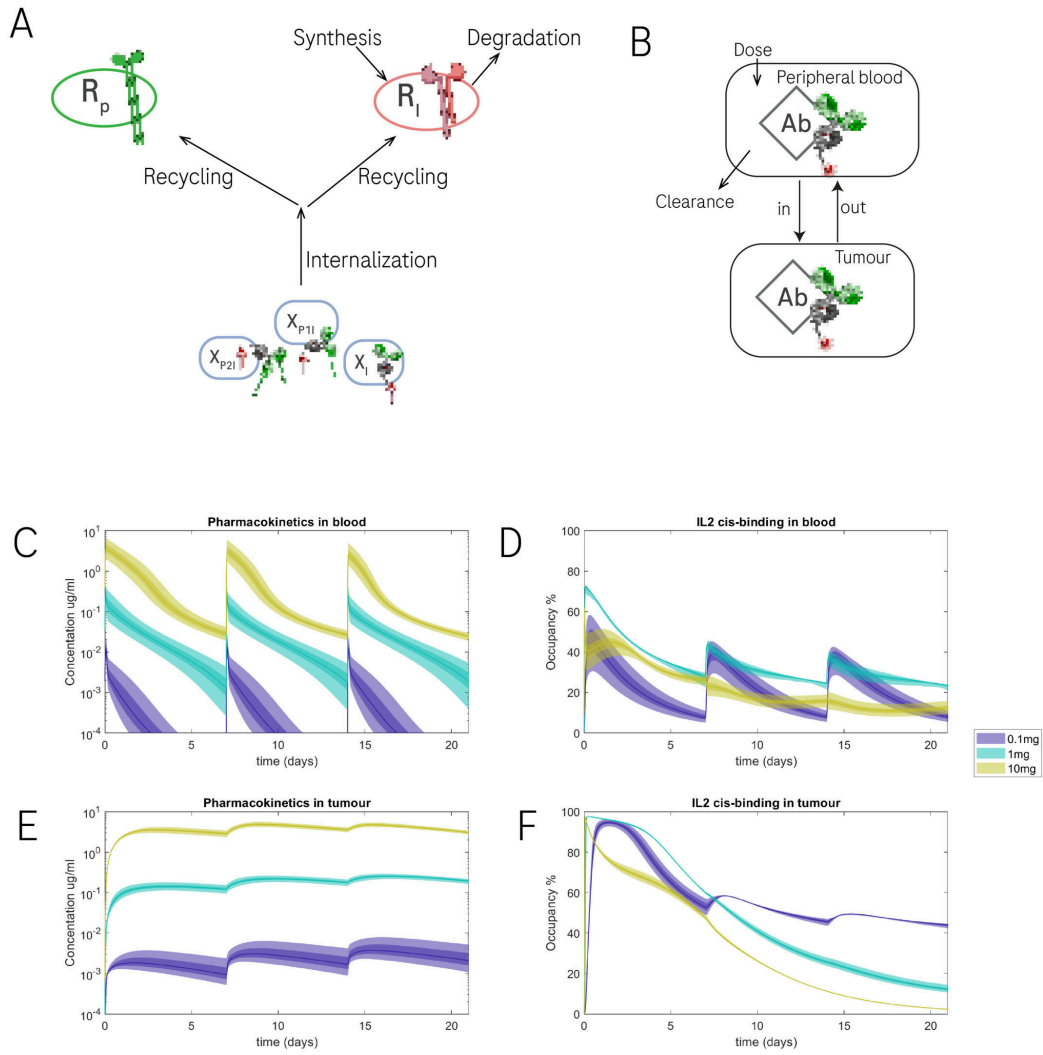


Figure 2: A: Schematic diagram illustrating the target mediated drug disposition extension to the binding model illustrated in Figure2: A. The model includes IL-2R turnover,

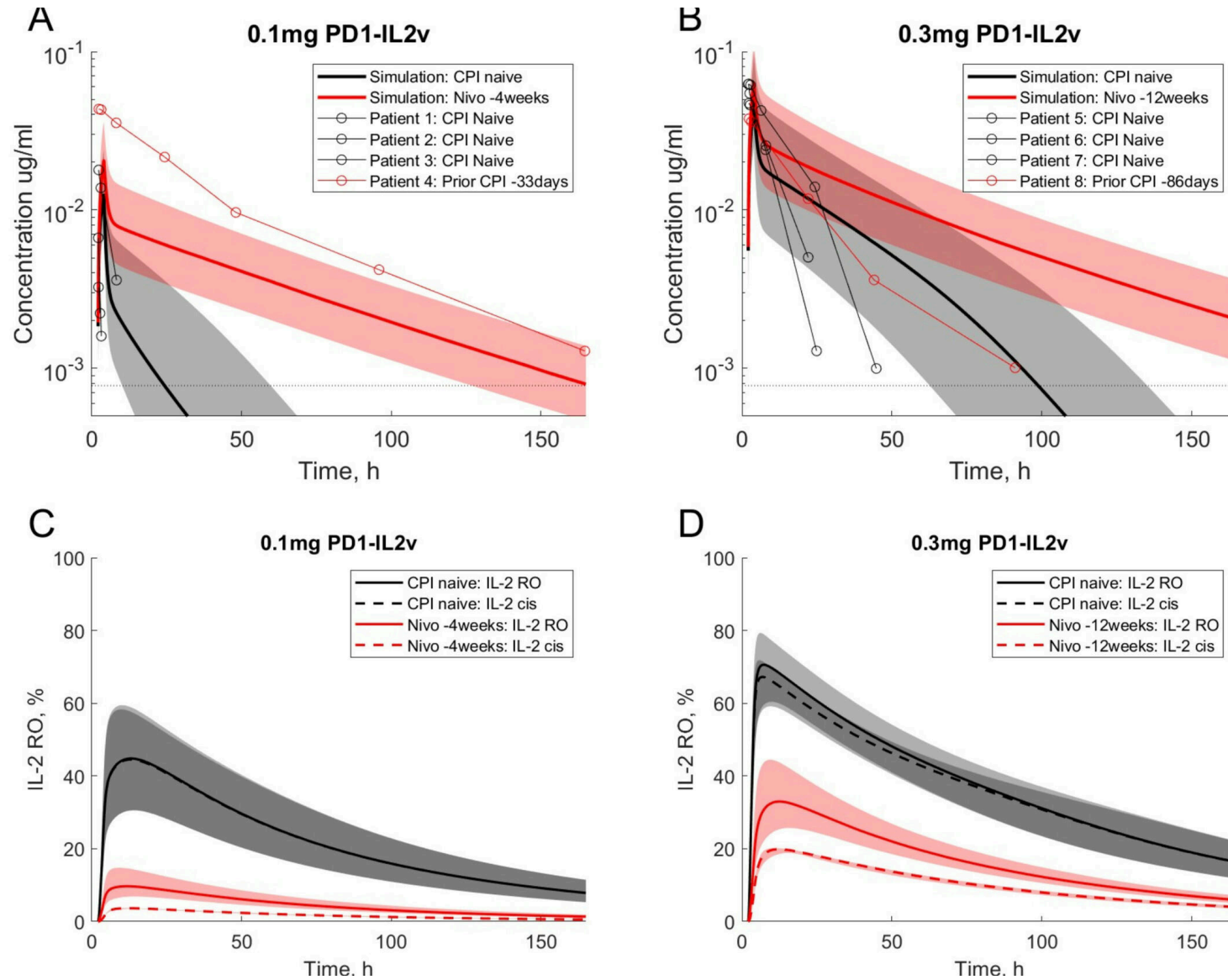


Figure 3: (A, B) Observed concentration-time profiles from the first cycle of treatment of PD1-IL2v from patients receiving 0.1 mg (A) or 0.3 mg (B) doses of PD1-IL2v show

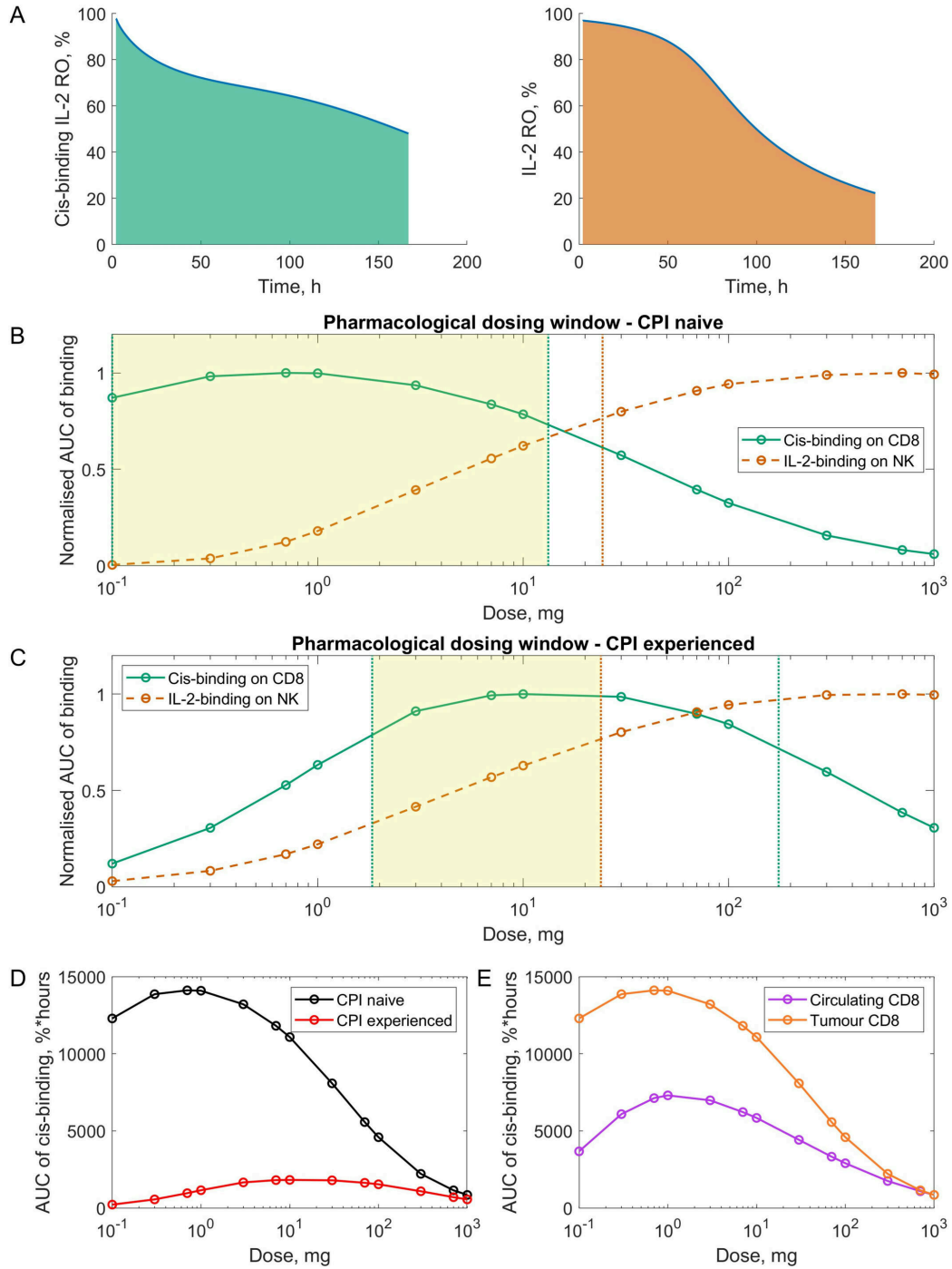


Figure 4: (A) Plots illustrating the calculation of the metrics related to the AUC of cis-bound IL-2 receptor occupancy on CD8 cells in the tumor (green shaded area) and t